

**CHARACTERIZATION OF AZ91 METAL MATRIX COMPOSITE
REINFORCED GRAPHENE**

NUR HAZIMAH BINTI SADON

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This Final Year Project Report entitled “**Characterization of AZ91 Metal Matrix Composite Reinforced Graphene**” was submitted by Nur Hazimah Binti Sadon in partial fulfilment of the requirements for the Degree of Bachelor of Science (Hons.) Physics, in the Faculty of Applied Sciences, and was approved by

Assoc. Prof. Dr. Saidatulakmar Binti Shamsuddin
Supervisor
B. Sc. (Hons.) Physics
Faculty of Applied Sciences
Universiti Teknologi MARA
02600 Arau
Perlis

Dr. Nur Hidayah Binti Ahmad Zaidi
Co-supervisor
School of Materials Engineering
Universiti Malaysia Perlis
02600 Arau, Perlis

Dr. Khuzaimah Binti Nazir
Project Coordinator
B. Sc. (Hons.) Physics
Faculty of Applied Sciences
Universiti Teknologi Mara
Perlis Branch, Arau Campus,
02600, Arau, Perlis

Dr Rosyaini Binti Afindi Zaman
Programme Coordinator
B. Sc. (Hons.) Physics
Faculty of Applied Sciences
Universiti Teknologi Mara
Perlis Branch, Arau Campus,
02600, Arau, Perlis

Date: _____

AUTHOR'S DECLARATION

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Name of Student	:	Nur Hazimah Binti Sadon
Student I.D. Number	:	2020897994
Programme	:	Bachelor of Sciences (Hons.) Physics – AS203
Faculty	:	Applied Sciences
Thesis Title	:	Characterization Of AZ91 Metal Matrix Composite Reinforced Graphene
Signature of Student	:	
Date	:	August 2023

ABSTRACT

CHARACTERIZATION OF AZ91 METAL MATRIX COMPOSITE REINFORCED GRAPHENE

AZ91 is an aluminum-zinc magnesium alloy with high specific strength and good castability. It is commonly used in automotive and biomedical applications. However, AZ91 has relatively poor mechanical properties, such as low hardness and high porosity. This study aims to fabricate a metal matrix composite of AZ91 with Graphene using Powder Metallurgy method thus to determine the density, porosity, and hardness of the composite. The fabrication process involves mixing, compacting, and sintering. Graphene is a two-dimensional material with excellent mechanical, electrical, and thermal properties. It has been shown to be an effective reinforcement for metal matrix composites. Experimental analysis reveals that temperature significantly affects the properties of the composite, with hardness and density increasing while porosity decreases. The optimal sintering temperature is found to be 400°C. This research contributes to the development of high-performance materials, particularly for biomedical applications.

ABSTRAK

PENCIRIAN KOMPOSIT MATRIKS LOGAM AZ91 DIPERKUATKAN GRAFIN

AZ91 adalah aloi aluminium-zink magnesium dengan kekuatan spesifik yang tinggi dan kebolehcorak yang baik. Ia biasanya digunakan dalam aplikasi automotif dan bioperubatan. Walaubagaimanapun, AZ91 mempunyai sifat mekanikal yang agak lemah, seperti kekerasan rendah dan keporosan yang tinggi. Kajian ini bertujuan untuk menghasilkan komposit matriks logam AZ91 dengan Grafen menggunakan kaedah Metalurgi Serbuk dan seterusnya untuk menentukan ketumpatan, keporosan, dan kekerasan komposit. Proses pembuatan melibatkan pencampuran, pemampatan, dan pangsinteran. Grafen adalah bahan dua dimensi dengan sifat mekanikal, elektrik, dan terma yang cemerlang. Ia telah terbukti sebagai penguatan yang berkesan untuk komposit matriks logam. Analisis eksperimen mendapati bahawa suhu mempengaruhi sifat komposit secara signifikan, dengan kekerasan dan ketumpatan meningkat sementara keporosan berkurangan. Suhu pangsinteran optimal didapati pada 400°C. Kajian ini memberi sumbangan kepada pembangunan bahan berkualiti tinggi, terutamanya untuk aplikasi bioperubatan.

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(Nur Hazimah Binti Sadon)

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LIST OF SYMBOLS

%	Percentage
wt. %	Weight Percent
°C	Degree Celcius
ρ	Density
α	Alpha
β	Beta

LIST OF ABBREVIATIONS

Al	Aluminium
AE21 element	Composition of element Al 9%, Zn 1% others are Mg
AZ31 element	Composition of element Al 3%, Zn 1% others are Mg
AZ91 element	Composition of element Al 9%, Zn 1% others are Mg
CNTs	Carbon nanotubes
et al.	And others
GRMMCs	Graphene reinforced metal matrix composites
GPa	Giga pascal
HrF	Rockwell Hardness Scale F(1/16 mm ball diameter)
Mg	Magnesium
MMC	Metal Matrix Composite
PM	Powder Metallurgy
rpm	Rotation per minute
Ti	Titanium
XRD	X-ray Diffraction
Zn	Zinc

CHAPTER 1

INTRODUCTION

1.1 Background Of Study

AZ91 is a magnesium alloy in the Mg-Al-Zn series that comprise 9 wt % Al and 1 wt % Zn. Currently, it is the most prevalent and frequently utilised in the Mg alloy series (Zhao et al., 2022). AZ91 has a wide range of applications in the aerospace and automotive industries, including chassis, indoor frames, and wheels. (Mathiazhagan & Jayabharathy, n.d.). Additionally, AZ91 has been used as a bone implant in the biomedical sector. AZ91 has high strength to weight ratio and high specific modulus (L et al., 2020). In contrast, AZ91 has poor corrosion resistance and low wear resistance in some environments. (Mathiazhagan & Jayabharathy, n.d.). AZ91 also lose strength above 120 softening of the β -phase which restricts its use in some environments. (L et al., 2020). Reinforcement of AZ91 magnesium alloy with appropriate material such as Graphene, Carbon Nanotube, Titanium and Titanium Dioxide can overcome the alloy's shortcoming. Graphene has been chosen in this study as reinforcement due to their good properties.

The addition of graphene improves the grain refinement of composite materials, reducing porosity and improving fluidity. A unique structural feature of graphene also contributes to its high yield strength, tensile strength, and Young's modulus (Zhao et al., 2022). In this study, powder metallurgy method will be used to fabricate a metal matrix composite of AZ91 magnesium alloy reinforced with graphene.

1.2 Problem Statement

AZ91 alloy has attracted the attention of many industries, including automotive as a steering, wheel. In biomedical field as a bone implant. Unfortunately, AZ91 lose strength above 120 softening of the β -phase, limiting its use in some environments. This issue must be resolved before the use of AZ91 can be expanded. Therefore, there are a few things that can be done to solve the issue, such as the addition of the proper alloying elements, reinforcement, and any other surface modification.

Next, AZ91 also corrodes easily in in the majority of aqueous environments, including the fluid found in human bodies and this could result in an unintended early failure of the implant. To ensure that AZ91 can be used as a reliable implant in biomedical industry, this issue needs to be fixed by adding some reinforcement materials. Graphene will be used to reinforce AZ91 by powder metallurgy in order to overcome these restrictions.

During fabrication, the sintering temperature must be precisely controlled. The sintering temperature has a significant impact on the densification and microstructure growth of graphene-reinforced AZ91. Adjusting the sintering temperature improves grain refinement, particle distribution, and interfacial bonding between the graphene reinforcement and the AZ91 matrix. It is expected that by systematically studying different sintering conditions, an optimal temperature range that can effectively address the strength loss and corrosion susceptibility issues will be identified.

Some research question involved in this study:

- i. How can powder metallurgy help in dispersion rate of Graphene?

According to the findings of an experiment conducted by El-Kady et al., (2022), powder metallurgy can aid in the dispersion rate of graphene. Excellent adhesion and homogeneous dispersion of the investigated reinforcement were achieved using powder metallurgy. (El-Kady et al., 2022).

ii. Why Graphene had been chosen instead of CNT?

Graphene had been chosen due to its advantages. The primary advantage of graphene over carbon nanotubes is that it has a larger specific surface area (Prashantha Kumar & Anthony Xavier, 2014).

1.3 Significance of study

In light of this study, powder metallurgy may be used to create a new, cost-effective product. This is due to the fact that only two to four materials are required to make a new product. The fabrication of novel, inexpensive products will subsequently benefit most sector such as biomedical and automotive sectors.

1.4 Objectives Of Study

1. To fabricate a metal matrix composite of AZ91 reinforced with Graphene.
2. To determine physical and mechanical of metal matrix composite AZ91 reinforced with Graphene.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction to composite

Composites can be found in a wide range of appliances today, from home construction to engineering. Composite materials are a combination of two or more materials with different physical and chemical properties that will produce new products. Composites can be classified into three major types which are polymer matrix, metal matrix, and ceramic matrix (Mathiazhagan & Jayabharathy, n.d.). A composite material called Metal Matrix Composite (MMC) has metallic matrix and dispersed fibres or particles. The main matrix materials used are aluminium, titanium, magnesium and copper (Mathiazhagan & Jayabharathy, n.d.).

2.2 Magnesium

Magnesium (Mg) has a density of 1.738g/cm^3 , which is nearly one-fourth that of steel and two-thirds that of aluminium. The atomic number of Mg is 12 and the molar mass is 24 element. Mg is a shiny. Despite the fact that magnesium has a very high effectiveness, it cannot be used directly in many applications due to its high reactivity, low melting point of 650 degrees, high corrosion rate, high creep rate, and high solidification shrinkage of about 4.2% .

Table 2.1 Properties of Mg

Properties	Magnesium
Atomic number	12
Atomic weight	24.32
Crystal structure	HCP
Density	1.74
Melting point	650
Boiling point	1105
Thermal expansion coefficient	25.5

Source: Sivamaran (2022)

2.2.1 Magnesium alloy

Many magnesium alloys with different metal additions, such as aluminium, zinc, zirconium, manganese, and rare earth elements, have been developed to overcome the problem associated with pure Mg (L et al., 2020). Magnesium alloys are alloys composed of magnesium (the lightest structural metal) and other metals, most commonly aluminium, zinc, manganese, silicon, copper, rare earths, and zirconium. Magnesium alloys have a hexagonal lattice structure, which influences their fundamental properties (Khademian & Peimaei, 2021). When a metal is alloyed, its properties such as strength, hardness, and corrosion resistance change. Mg-Al series alloys, such as the magnesium alloy AZ91, are now widely used (Tan et al., 2012).

2.2.2 AZ91

AZ91 is one of the most commonly used magnesium alloy with density of 1.82 g/cm³ (Zhao et al., 2022). This alloy contains 9 wt % Al and 1 wt % Zn (L et al., 2020). AZ91's tensile strength is 160-365 MPa and elastic modulus is 45 GPa are two-thirds that of aluminium (Čížek et al., 2004).

The advantages of AZ91 are high specific strength, low cost, good castability (Zhao et al., 2022). According to (Mathiazhagan & Jayabharathy, n.d.), AZ91 is less prone to hydrogen porosity and has less reactive iron that can be poured into steel moulds. AZ91 alloy also has a high strength-to-weight ratio (tensile strength/density) that is comparable to other structural metals (Čížek et al., 2004). When compared to monolithic magnesium, the AZ91 has far superior properties (Velmurugan et al., 2020). Because of its advantages, AZ91 is an excellent choice for use in a variety of applications, including the aerospace and automotive industries. (Čížek et al., 2004). For example, car seat frames, steering, wheels and laptop, gear boxes, air intake system and many more (Zhao et al., 2022).

Limited mechanical strength and poor friction properties are two drawbacks of AZ91 (Zhao et al., 2022). (L et al., 2020) claims that above 120 degrees, it starts to lose strength and other properties. This happens as a result of the β -phase softening (L et al., 2020). In some service environments, AZ91 also has poor resistance and low wear resistance (Mathiazhagan & Jayabharathy, n.d.).

Table 2.2 Chemical composition of AZ91 alloy powder

Element	Al	Zn	Mn	Si	Cu	Fe	Ni	Be
Content (wt%)	9.08	0.65	0.23	0.038	0.003	0.0021	0.00098	0.00082

Sources: Zhao et al., (2022)

2.3 Reinforcement

Reinforcement is combination of a composite material that will enhance desired properties of the conventional material depend upon the working environment in which the material is exposed such as strength, stiffness, corrosion resistance, wear resistance and many more (Mathiazhagan & Jayabharathy, n.d.). Reinforcement refers to a material that has a good bonding and strengthens the matrix material (Selvam et al., 2021). The reinforcement can be extremely hard and robust while remaining extremely light in weight (Selvam et al., 2021). They are divided into categories like short fibres, whisker, particulates, and preform/wire(fibre solid) (Mathiazhagan & Jayabharathy, n.d.).

2.3.1 Selection of Reinforcement

Selection of reinforcement plays a crucial role as improving the desired properties of final material. The following factors must be considered when selecting reinforcement materials such as size, diameter, aspect ratio, surface morphology, voids, inherent properties, such as strength, moduli, and density, and chemical compatibility with the matrix. (Kapranos et al., 2014). Some reinforcements that are always been utilised to reinforce AZ91 are Graphene, Carbon Nanotube, Titanium and Titanium Dioxide.

2.3.2 Graphene

Graphene has impressed us in recent years due to its superior mechanical properties (Yuwaraj et al., 2021). It consists of a single basic layer of a two-dimensional lattice assembly made of sp²-hybridized carbon atoms (a two-dimensional honeycomb structure) (Tabandeh-Khorshid et al., 2020). In other words, graphene is the name given to a single hexagonal layer of graphite (Guler et al., 2020).

Graphene is a superior material that is a good candidate for use as a reinforcing material in many metals, polymers, and even ceramic matrix composites due to its extremely high surface area, excellent thermal conductivity, high electron mobility, and mechanical strength beyond that of conventional materials (Guler et al., 2020). Given its high surface area, low density of 1 g/cm³, and 130 Gpa Young's modulus, graphene has the potential to be an excellent reinforcement for metal matrix composites (Tabandeh-Khorshid et al., 2020). The primary advantage of using graphene over CNTs is that it has a larger specific surface area (Prashantha Kumar & Anthony Xavier, 2014).

2.4 AZ91 Reinforced with Graphene

Graphene is the novel reinforcement of AZ91 due to their unique mechanical and physical properties such as an increase in ultimate tensile strength to 190 MPa, which is 1.32 times greater than unreinforced AZ91 (Kandemir, 2018) but Graphene is more difficult to disperse than other reinforcements due to its larger interfacial contact area (Tabandeh-Khorshid et al., 2020).

2.5 Fabrication of AZ91 reinforced Graphene

2.5.1 Powder metallurgy

Powder metallurgy is a manufacturing method that uses powder metal in today's developing industry. Powder metallurgy, an advanced manufacturing method, allows for the production of all technological parts (Beştekin, 2010). According to R.Raja & M.Rajkumar (2015), there are many different definitions of PM, some of which include: Science of producing metal powders and making finished or semi-finished objects from mixed or alloyed powders with or without the addition of nonmetallic constituents, Powder metallurgy is the process of creating components with the necessary properties and shape by combining metal and non-metal powders. It refers to the preparation and processing of powdered iron and nonferrous metal. By combining metal and non-metal powders, powder metallurgy (PM) creates components with the necessary characteristics and shape.

The benefits of powder metallurgy include no material is wasted while obtaining high dimensional accuracy, shape, and surface finish. The product can be produced with the required density and porosity, and it can be produced at a high rate while maintaining the required properties. This process uses all of the material available (R.Raja & M.Rajkumar, 2015). According to Beşteş, (2010), PM also enables the production of highly mixed material alloy combinations which are not possible to make using other processes. Due to the benefits, simplicity, and stability of this method, many industries have utilised it extensively in their production. This method, which is essential for large sectors, is used in the following industries: Aerospace, abrasion resistance, fuel cells, and material processing (Beşteş, 2010).

The disadvantages of PM are that it is not cost effective for small-scale production. The cost of equipment and dies is prohibitively expensive. Large components cannot be manufactured. The density of the component is not uniform. When powder is stored, it can deteriorate due to oxidation, and broken parts cannot be rejoined (R.Raja & M.Rajkumar, 2015).

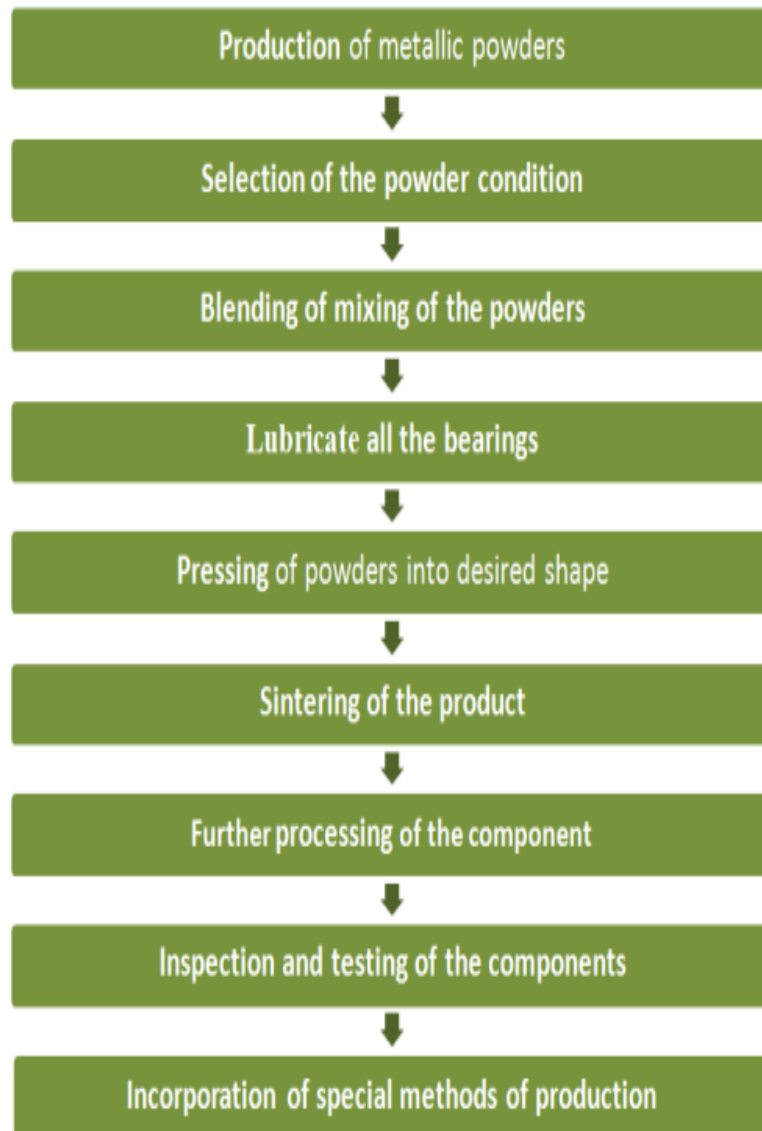


Figure 2.1 Steps involved in process of powder metallurgy

Source: (R.Raja & M.Rajkumar, 2015)

2.5.2 Mixing of Raw

The mixing step, according to (Beştekin, 2010), is the step in which the raw powder is combined with some alloy elements, lubricant, or special additives until it is homogeneous.

2.5.3 Compaction

Compaction is the step that involves forming loose powders into required shapes with sufficient strength to withstand until sintering is completed (Youness & Taha, 2021). Mechanical or hydraulic pressures are used to shape (Beştekin, 2010). The most important factor to consider is compression pressure. Insufficient compression will result in weak bonding between the reinforcement and matrix. If the pressure is too high, the sample will collapse and the structure of the space holder will be destroyed.

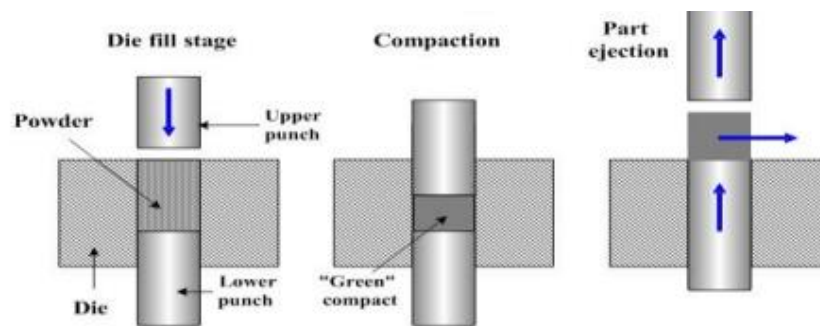


Figure 2.2 Powder Compaction

Source:(Umamaheshwer Rao et al., 2018)

2.5.4 Sintering

Sintering is the process of heating compacted powder to a specific temperature that is below the main powder particles' melting point but above the necessary temperature to allow diffusion between nearby particles. (Youness & Taha, 2021).

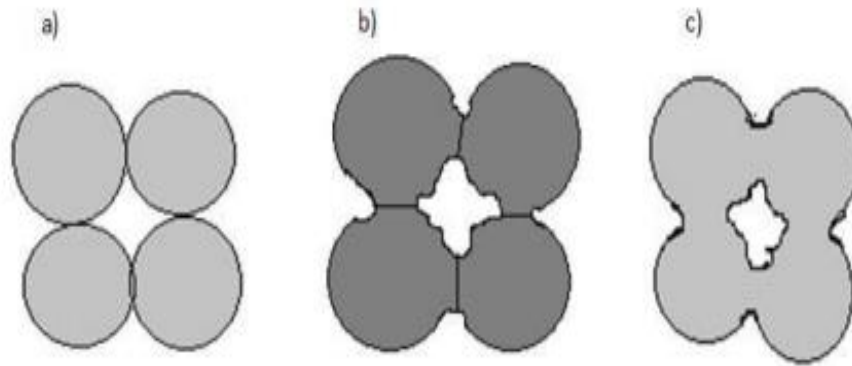


Figure 2.3 Illustration of Sintering

Source:(Umamaheshwer Rao et al., 2018)

2.6 X-ray Diffraction (XRD)

X-ray diffraction (XRD) is a highly effective nondestructive method for characterising crystalline materials. Structures, phases, preferred crystal orientations (texture), and other structural parameters such as average grain size, crystallinity, strain, and crystal defects are all provided. Constructive interference of a monochromatic beam of X-rays scattered at specific angles from each set of lattice planes in a sample generates X-ray diffraction peaks. The atom distribution within the lattice determines the peak intensities.

The benefits of XRD include its potency and speed (20 min) for identifying unknown minerals, its ability to reliably determine a mineral's identity in the majority of cases, its minimal sample preparation requirements, the accessibility of XRD units across a wide range of applications, and its relatively simple data interpretation (Bunaciu et al., 2015).

The disadvantages of XRD are that access to a standard reference file of inorganic compounds is required, material, in tenths of a gramme quantity, must be ground into a powder, for mixed materials, detection limit is » 2% of sample, for unit cell determinations, indexing of patterns for nonisometric crystal systems is complicated, and peak overlay may occur and worsens for high angle "reflections" (Bunaciu et al., 2015).

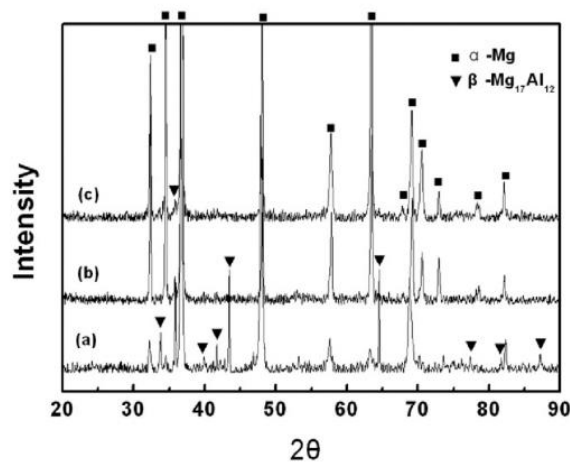


Figure 2.4 XRD patterns of AZ91D magnesium alloy

Source: (Chai et al., 2015)

2.7 Rockwell Hardness Test

Rockwell hardness testing is a method of determining the hardness of a specimen. It assesses a specimen's resistance to indentation under constant load for a short period of time (L et al., 2020).

Rockwell hardness testing is distinct from other hardness tests. Although the fundamentals remain the same, the procedures used in Rockwell hardness testing are more systematic, allowing measurements to be taken quickly, easily, and with excellent reproducibility. Rockwell hardness testing machines are also user-friendly, with the hardness number read directly from a dial (Yazdanshenas & Goh, 2020).

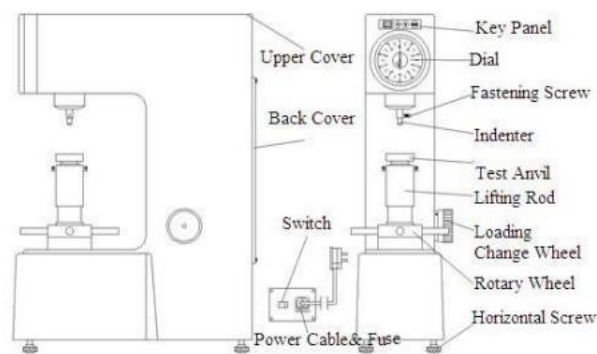


Figure 2.5 Rockwell Hardness Test Machine Schematic

Source : Yazdanshenas & Goh (2020)

2.8 Properties AZ91 Reinforced with Graphene

Based on the results of the experiment by Yuwaraj et al., (2021), metal matrix composite of AZ91 reinforced with graphene has better mechanical and physical properties. Some test has been conduct in order to determine the physical and mechanical properties of metal matrix composite of AZ91 reinforced with Graphene such as porosity and density test. Metallographic of AZ91 reinforced with Graphene also were determined. The hardness of the materials also were determined.

2.8.1 Density and porosity

The material's density and porosity are critical to analyse because they have a large impact on its application. Density is defined as the mass per unit volume, whereas porosity is the measure of void space in a material. According to a study published in the (Zhao et al., 2022), adding graphene reduces porosity.

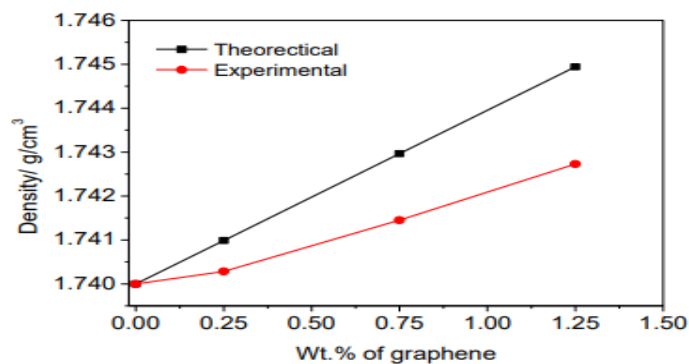


Figure 2.6 The effect of wt.% of graphene on density

Source: DU & , K. F. ZHEN (2018)

2.8.2 Metallographic analysis

Metallographic analysis is important in order to determine how the material will perform in a given application. The distribution of reinforced particles in matrix material was investigated using metallographic analysis (Jayabharathy & Mathiazhagan, 2019). According to Zhao et al., (2022), the addition of Graphene have significant effect on grain refinement of composite materials.

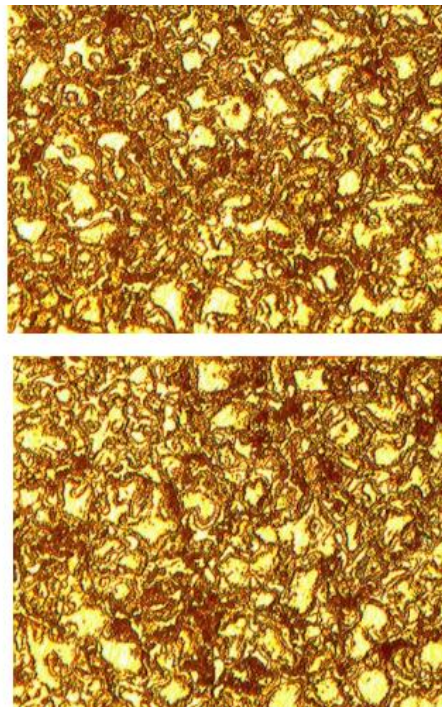


Figure 2.7 Micro-structure of AZ91 with 0.3 wt% of Graphene
Source: Yuwaraj et al., (2021)

2.8.3 Hardness

Hardness of materials correlates directly with its strength, ductility and wear resistance. Based on the result of the experiment that had been conducted by (Yuwaraj et al., 2021), the hardness of AZ91 increased as the amount of graphene increased.

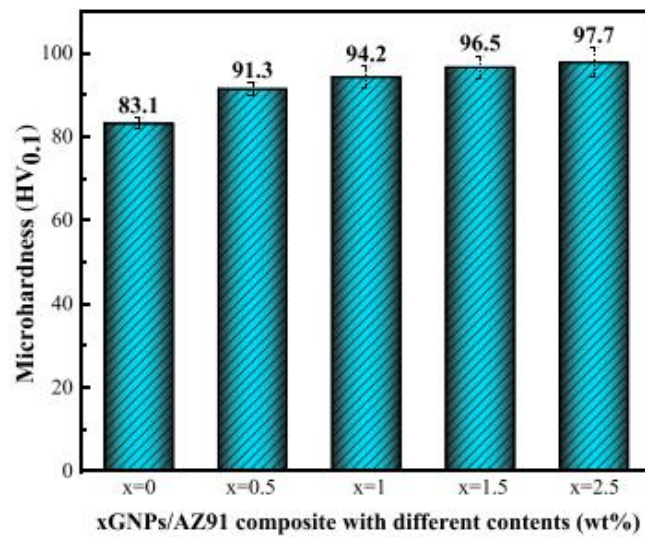


Figure 2.8 Microhardness of AZ91 alloy and xGNPs/AZ91 composite with different contents ($x = 0.5, 1.0, 1.5$, and 2.5 wt%)

Source: Zhao et al., (2022)

2.9 Application of AZ91

2.9.1 Bone implant

In commercial Mg alloys, Al is the most added element, which not only refines the grains but also improves corrosion resistance (Zhang et al., 2022). The Al-rich layer and β -phase ($Mg_{17}Al_{12}$) network formed on the corrosion surface after the addition of Al can effectively prevent further corrosion of Mg alloys (Zhang et al., 2022). Because of their good mechanical properties and corrosion resistance, many Mg alloys containing Al element, such as AE21, AZ31, and AZ91, have been used in biomedical fields (Zhang et al., 2022).

Numerous factors, such as the stiffness, strength, and toxicity of the materials, should be considered when designing an excellent bone implant because they will come into contact with living things. Mg alloy is a novel medical material that can replace conventional bone implants like steel nails as an orthopaedic implant because it more closely resembles the mechanical characteristics of human bone and doesn't have the same stress shielding effects as Ti alloys (Zhang et al., 2022).

According to (Sivamaran, 2022), Magnesium and human bones share the same density. Furthermore, after the human bones have mostly healed, Mg alloys can be degraded into non-toxic and harmless small

molecules and excreted through the human circulatory system, sparing patients the pain of a second surgery to remove the implant (Zhang et al., 2022).

2.9.2 Automotive

In recent years, AZ series casting Magnesium alloys, especially AZ91 alloy, have been used in some structural components of automobiles and aircraft for reasons such as high specific strength, lightness, environmental friendliness, conductivity. Volkswagen, one of the first to use these alloys, used approximately 22 kg of magnesium alloy in the Beetle model vehicle (A.H & A.G ,2012).

By additional use of this low-density magnesium alloys will receive higher benefits in the automotive industry. Some of the automotive applications of magnesium alloys are AZ91D (Mg-9Al-1Zn) which is widely used in the automotive industry because of its better corrosion resistance, superb die-cast ability, and fit for room temperature. So, this alloy is used as crankcase, steering box, steering column, brake pedal, manual transmission case, drive brackets and sub frames, etc. (Sivamaran, 2022).

CHAPTER 3

METHODOLOGY

3.1. Introduction

In this chapter, the powder metallurgy method employed to fabricate AZ91 reinforced with graphene is thoroughly described. The sample's structure is characterized using X-ray Diffraction (XRD), shedding light on its composition. The metallographic properties of the sample are carefully examined through the utilization of an optical microscope, enabling a detailed assessment. The density and porosity of the sample are determined by employing the Archimedes principle, revealing crucial material attributes. The hardness of the composite is measured using the Rockwell hardness test, contributing to a comprehensive understanding. Furthermore, a visual representation, Figure 3.1, portrays a flowchart outlining the step-by-step process employed in fabricating AZ91 reinforced graphene. As this chapter unfolds, readers will develop a more profound grasp of the methodical approach utilized to merge AZ91 and graphene, showcasing the harmonious blend of scientific investigation and hands-on execution.

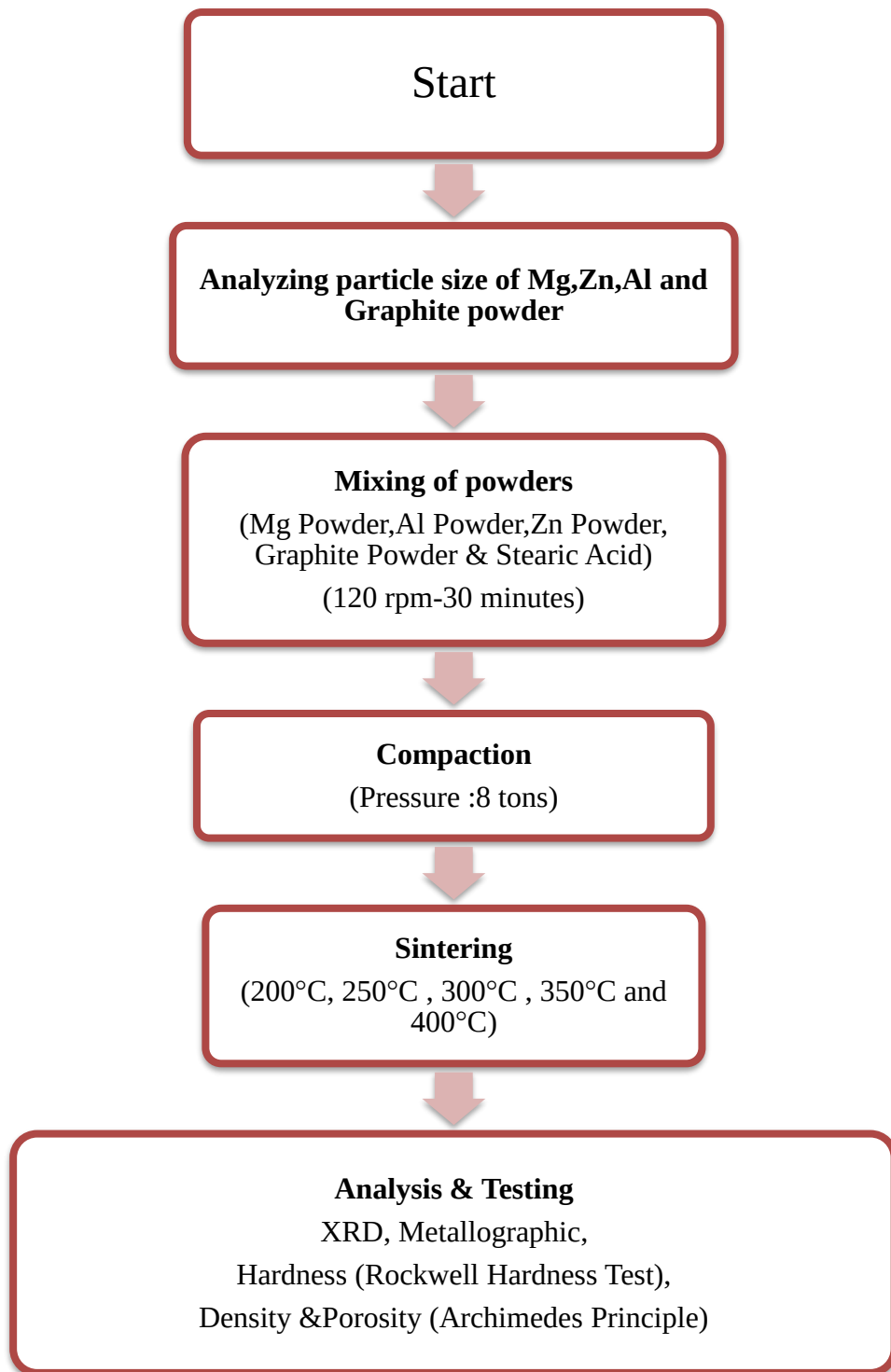


Figure 3.1 Flowchart of overall process to fabricate AZ91 reinforced with Graphene

3.2. Raw Material

Magnesium (90 wt %) powder, Aluminium powder (9 wt %), Zinc powder (1 wt%), Graphite powder (5 wt%) and stearic acid (3 wt %) were used as raw materials in this experiment. The calculation for the amount of raw materials that were used shown in the table 3.1:

Amount of Raw material = weightage percentage x 2g x total number of sample **Equation 3.1**

Table 3.1 : Amount of Raw Materials

Raw Material	Weightage percentage $\times$ 2 grams $\times$ Total number of sample	Amount of Raw Material
Mg Powder	$0.9 \times 2 \text{ grams} \times 5 \text{ samples}$	9 grams
Al Powder	$0.09 \times 2 \text{ grams} \times 5 \text{ samples}$	0.9 grams
Zn Powder	$0.01 \times 2 \text{ grams} \times 5 \text{ samples}$	0.1 grams
Graphite Powder	$0.05 \times 2 \text{ grams} \times 5 \text{ samples}$	0.5 grams
Stearic Acid	$0.03 \times 2 \text{ grams} \times 5 \text{ samples}$	0.3 grams

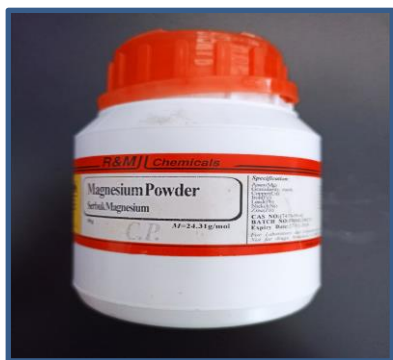


Figure 3.2(a) Magnesium Powder

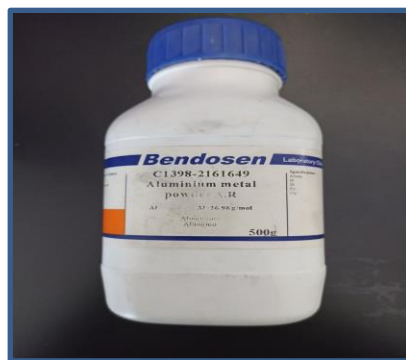


Figure 3.2(b) Aluminium Powder

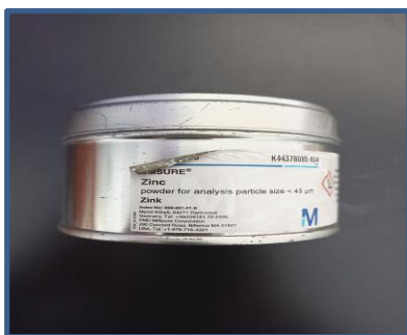


Figure 3.2(c) Zinc Powder

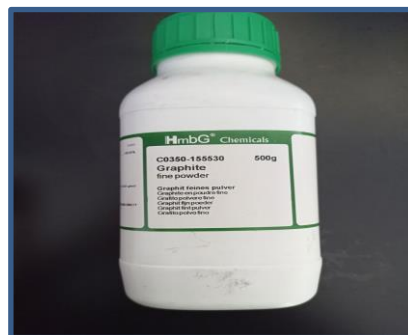


Figure 3.2(d) Graphene Powder

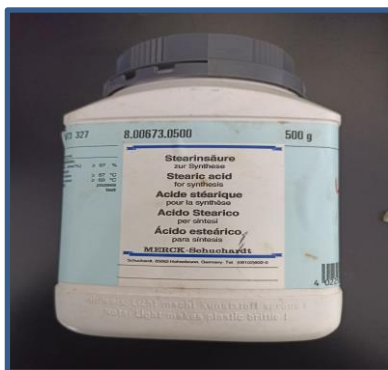


Figure 3.2(e) Stearic Acid

3.3. Particle Size

In this study, the distribution of particle sizes in our samples was carefully examined using a specialized tool. The comprehension of particle size distribution is of paramount importance as it affords insights into the composition, behavior, and properties of the material under investigation. By understanding particle size distribution, significant information about factors such as material performance, reactivity, and overall quality can be derived, allowing for informed decision-making and meaningful conclusions regarding the material's attributes. To ensure the precision of particle size analysis, a preparatory step involved gently passing the metal powders—specifically magnesium (Mg), zinc (Zn), aluminum (Al) and graphite—through a fine mesh featuring small 65-micrometer apertures. Following this, the powders were exposed to a laser beam, and the scattered light resulting from this interaction was measured from different angles. By analyzing the brightness and patterns within the scattered light, valuable insights into particle sizes were obtained. This analytical process encompassed the utilization of intelligent mathematical calculations based on the observed light patterns. The contribution of the tool enabled the precise determination of particle size distribution, ultimately providing valuable details about the composition and characteristics of the materials being examined.

3.4. Fabrication of AZ91 reinforced Graphene

AZ91 reinforced with Graphene was prepared by PM method that comprising some stages mixing, compaction and sintering. Sintering temperature was varied (200, 250, 300, 350, 400) °C to determined the optimum sintering temperature.

3.4.1. Mixing

The first step to prepare Magnesium AZ91 reinforced graphene is by mixing pure Mg powder with Al powder (9 wt %) , Zn powder (1 wt%) , Graphene (5 wt%) and stearic acid (3 wt%). The mixture sample was placed in a container. The container is then placed on a roller for approximately 30 minutes at 120 rpm. Figure below show the rotating mixer drum.



Figure 3.3 Panasonic DVUX960W rotating mixer drum

3.4.2. Compaction

For each sample, a mixture of magnesium, alumina, zinc, and graphene was weighed at 2g and introduced into the mold. The resulting mixture was then subjected to compaction using a Speacac Hydraulic Press at a pressure of 8 tons for a duration of two minutes. Prior to commencing the sintering process, the initial diameter and thickness of the green samples obtained after compaction were measured using a digital Vernier caliper. The laboratory hydraulic pressure is depicted in the accompanying figure.

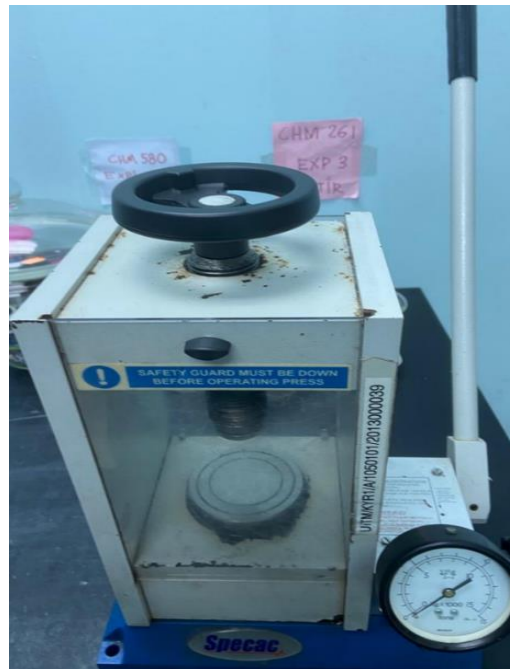


Figure 3.4 Speacac Hydraulic Press

3.4.3. Sintering

The obtained green body was sintered in a vacuum sintering furnace at five different temperatures: 200°C, 250°C, 300°C 350°C and 400°C for 2 hours for each sample. The heating rate was set up at 5 °C/minute. Figure below show one of the sintering profile of sample (at 400°C).

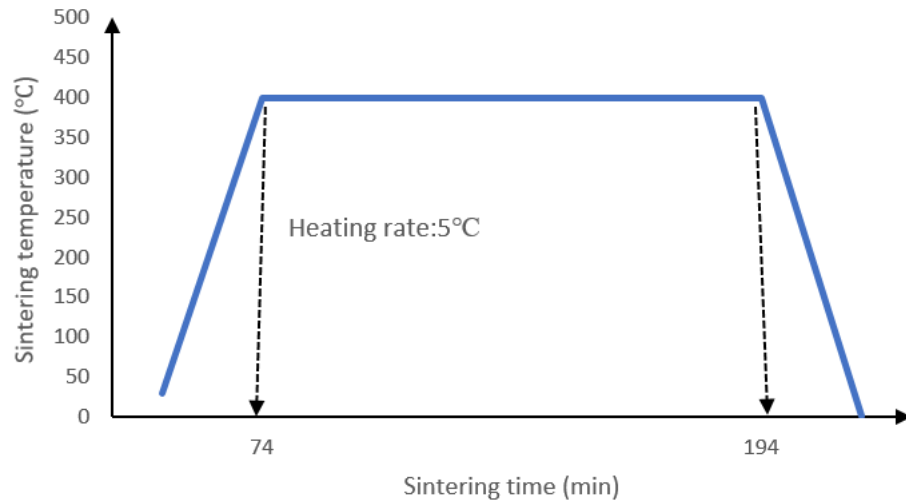


Figure 3.5 Sintering Profile

3.5. Characterization AZ91

This section look at mechanical and physical characteristics of metal matrix of AZ91 reinforced with Graphene like XRD, density and porosity, metallography and hardness.

3.5.1. XRD

X-ray Diffraction (XRD) is a diffraction pattern that demonstrates information about the crystal's structure or the behaviour of a crystalline substance as a result of the interference effect caused by the scattering of X-rays by a crystal's atoms. XRD was used to examine the elements in the sample. The elements observed using the XRD machine will then appear in HighScore Plus. HighScore Plus is a software suite for diffraction analysis. HighScorePlus software is depicted in Figure 3.6.

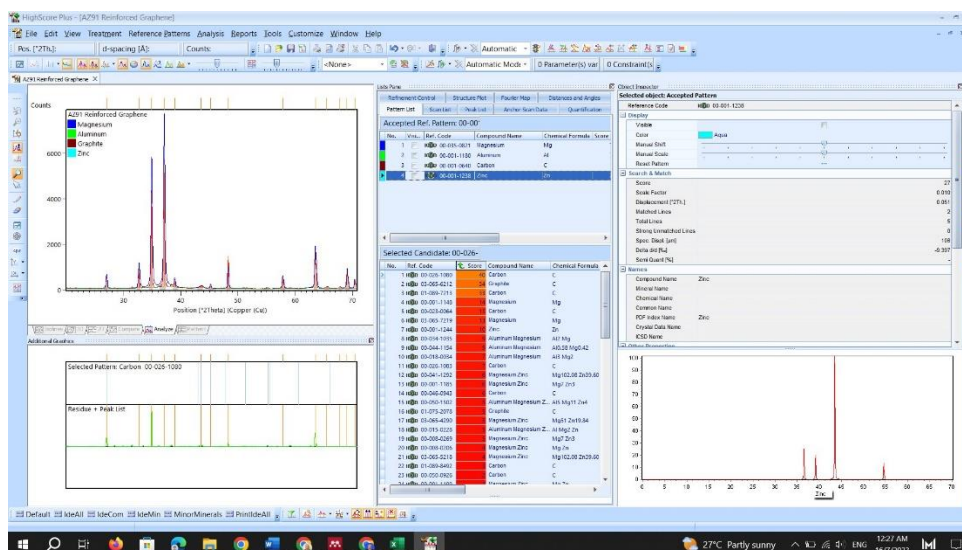


Figure 3.6 HighScorePlus software

3.5.2. Density and porosity

Archimedes principle was used to determine the value of density and porosity of metal matrix composite of AZ91 reinforced with Graphene. Densimeter had been used to calculate the density. The formula to calculate the density and porosity shown in equation 3.2 and 3.3. Figure 3.7 show the densimeter.

$$\text{Bulk Density} = \frac{\text{Weight of dry sample}}{W_c - W_b} \times \rho_{\text{water}} \quad \text{Equation 3.2}$$

$$\text{Relative Density} = \frac{\text{Bulk Density}}{\text{Theoretical Density}} \times 100\% \quad \text{Equation 3.3}$$

$$\text{Total Porosity} = 100\% - \text{Relative Density} \quad \text{Equation 3.4}$$



Figure 3.7 Densimeter

3.5.3. Metallographic analysis

ASTM E3-11, also known as the "Standard Practise for Preparation of Metallographic Specimens," is a widely accepted guideline developed by the American Society for Testing and Materials (ASTM). The methodology in this study followed the recommended procedures outlined in ASTM E3-11 for the preparation of metallographic specimens.

3.5.3.1. Mounting

The mounting procedure followed ASTM E3-11 guidelines. To embed the specimens, epoxy resin, specifically a 1:1 mixture of epoxy resin (normal) and epoxy hardener (crystal clear), was used. Prior to pouring the epoxy mixture into the moulds containing the AZ91 reinforced graphene samples, it was degassed in a vacuum desiccator to eliminate air bubbles and ensure a homogeneous mounting material. Figure 3.8 show the vacuum desiccator while figure 3.9 show the mounting process.



Figure 3.8 Vacuum Dessicator

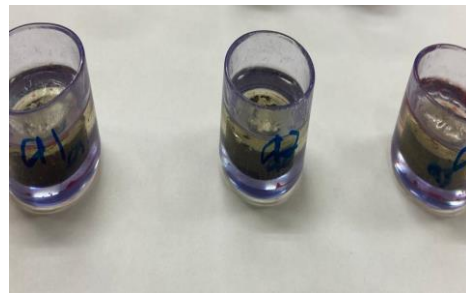


Figure 3.9 Process of Mounting Sample

3.5.3.2. Grinding and Polishing

The procedures for grinding were also in accordance with ASTM E3-11's guidelines. The samples were first ground with progressively finer grit sandpaper, with a range of 220 to 1000. A smooth and flat surface was intended to be created by removing material. Figure 3.10 show the grinding machine.



Figure 3.10 Grinding Machine

The polishing process was then carried out using diamond paste with progressively finer grit sizes of 6, 3, and 1 μm . In order to achieve a mirror-like finish—which is necessary for microscopic examination—this process further refined the surface.

3.5.3.3. Optical Microscope

The subsequent analysis was greatly aided by the optical microscope, which had 5x and 10x magnifications. The microscope made it possible to examine the fully ground and polished samples in detail and revealed the intricate surface morphology by taking high-resolution pictures. The sample that had been fully grind and polish was captured in order to study the surface morphology. Figure 3.11 show the optical microscope which was used for metallographic analysis



Figure 3.11 Olympus Optical Microscope

3.5.4. Rockwell Hardness Test

Figure 3.12 shows Rockwell Hardness test machine This machine is general method to measure the resistance of material to indentation. It is a widely used method for evaluating the bulk hardness of metallic and other materials.

Hardness test is carried out by using a 1/16 mm ball diameter, scale HrF, 588 N. Test was conducted on the sample and atleast at three point were measured to obtain the average value of hardness.



Figure 3.12 AFFRI Rockwell Hardness Test Machine

CHAPTER 4

RESULT AND DISCUSSION

4.1. Introduction

In this chapter, the effect of adding graphene to AZ91 with various temperature will be discussed. To initiate the analysis, the raw material, comprising AZ91 reinforced with graphene, will undergo characterization using X-ray diffraction (XRD). This technique will enable the examination of the crystal structure and composition of the material, providing valuable insights into its properties.

Subsequently, five different samples of magnesium AZ91 reinforced with graphene will be examined, with each sample being processed at a specific temperature. Through the utilization of a densimeter and Rockwell Hardness Test, essential properties such as density, porosity, and hardness will be measured. These measurements will offer a comprehensive understanding of the compactness of the material, the presence of voids, and its resistance to deformation.

4.2. Raw Material Characterization

4.2.1. Magnesium Powder

The particle size analyzer revealed that the particle size of Mg powder is 10% smaller than $32.717\ \mu\text{m}$ at $d(0.1) = 32.717\ \mu\text{m}$. The median particle size is represented by $d(0.5) = 65.363\ \mu\text{m}$ and the particle size is 90% smaller than $120.401\ \mu\text{m}$ at $d(0.9) = 120.401\ \mu\text{m}$. The results also show that Mg powder has a low specific surface area of $0.111\ \text{m}^2/\text{g}$. This indicates that the material has a small surface area, which could have an impact on its properties. The particle size distribution of Mg powder is shown in the graph below.

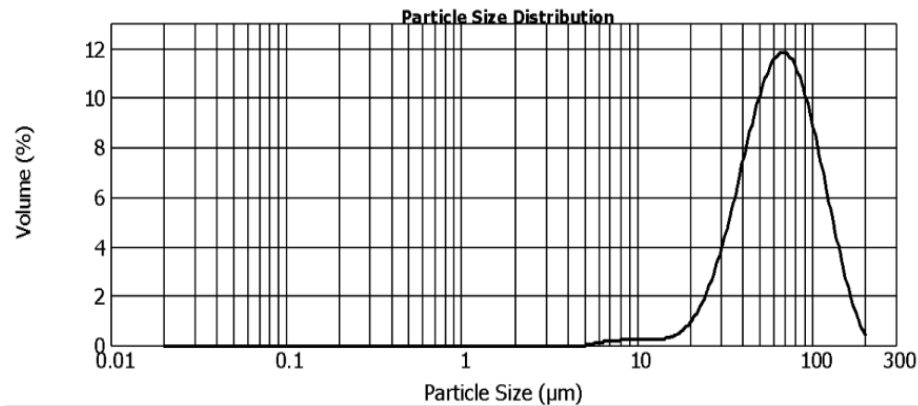


Figure 4.1 Particle Size Distribution of Mg

4.2.2. Aluminium Powder

In accordance with the results of the particle size analyzer, 10% of the particles in the sample are smaller than $9.108\ \mu\text{m}$ at $d(0.1) = 9.108\ \mu\text{m}$, 50% of the particles are smaller than $23.055\ \mu\text{m}$ at $d(0.5) = 23.055$, and 90% of the particles are smaller than $63.696\ \mu\text{m}$ at $d(0.9) = 63.696\ \mu\text{m}$. Furthermore, the results show that Al powder has a relatively high specific surface area of $0.336\ \text{m}^2/\text{g}$. This indicates that the material has a large surface area, which may affect its properties. The graph below depicts the particle size distribution of Al powder.

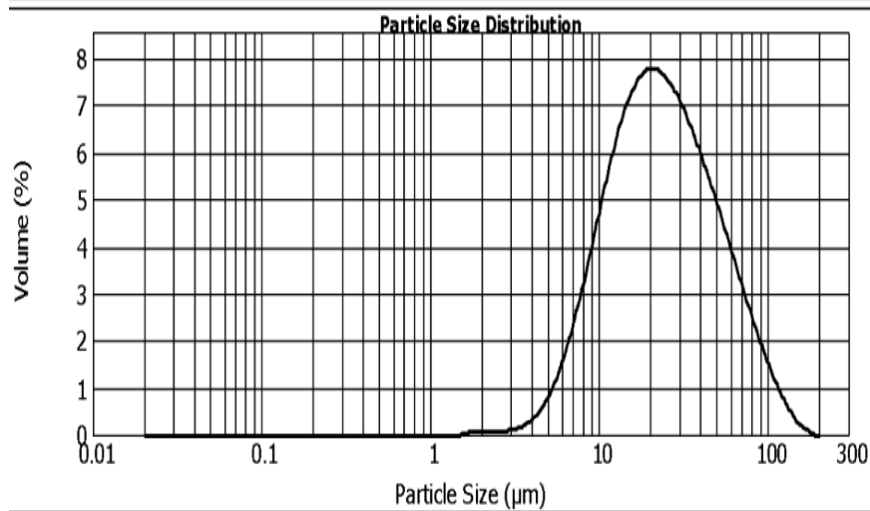


Figure 4.2 Particle Size Distribution of Al

4.2.3. Zinc Powder

The particle size analyzer demonstrated that at $d(0.1) = 6.342$, 10% of the particles in the sample are smaller than 6.342, at $d(0.5) = 12.972$, 50% of the particles in the sample are smaller than 12.972, and at $d(0.9) = 26.003$, 90% of the particles in the sample are smaller than 26.003. Furthermore, the results show that Zn powder has a relatively high specific surface area of $0.539 \text{ m}^2/\text{g}$. This indicates that the material has a large surface area, which may affect its properties. The graph below depicts the particle size distribution of Zn powder.

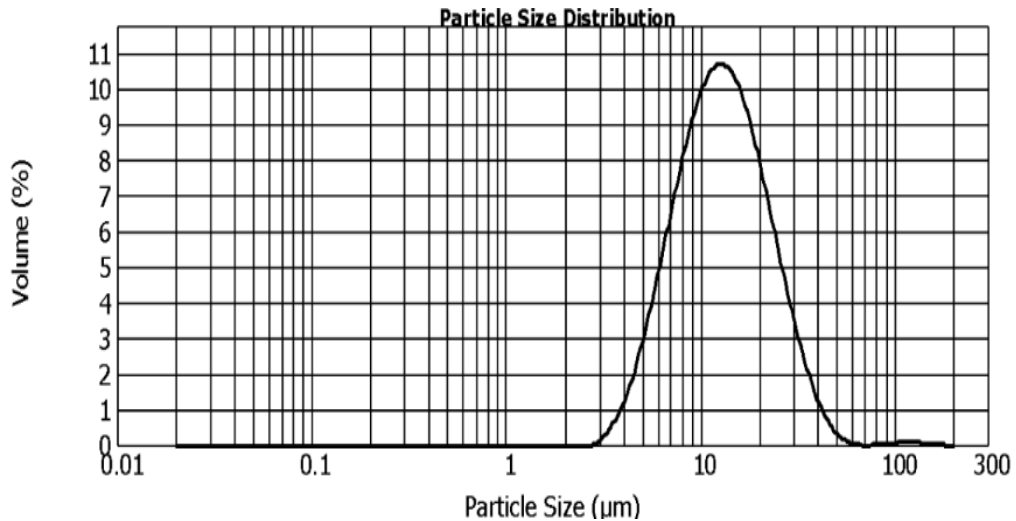


Figure 4.3 Particle Size Distribution of Zn

4.2.4. Graphene

The particle size analyzer demonstrated that at $d(0.1) = 12.795$, 10% of the particles in the sample are smaller than 12.795, at $d(0.5) = 42.635$, 50% of the particles in the sample are smaller than 42.635, and at $d(0.9) = 95.500$, 90% of the particles in the sample are smaller than 95.500. Furthermore, the results show that Graphene powder has a relatively high specific surface area of $0.239 \text{ m}^2/\text{g}$. This indicates that the material has a large surface area, which may affect its properties. The graph below depicts the particle size distribution of Graphene powder.

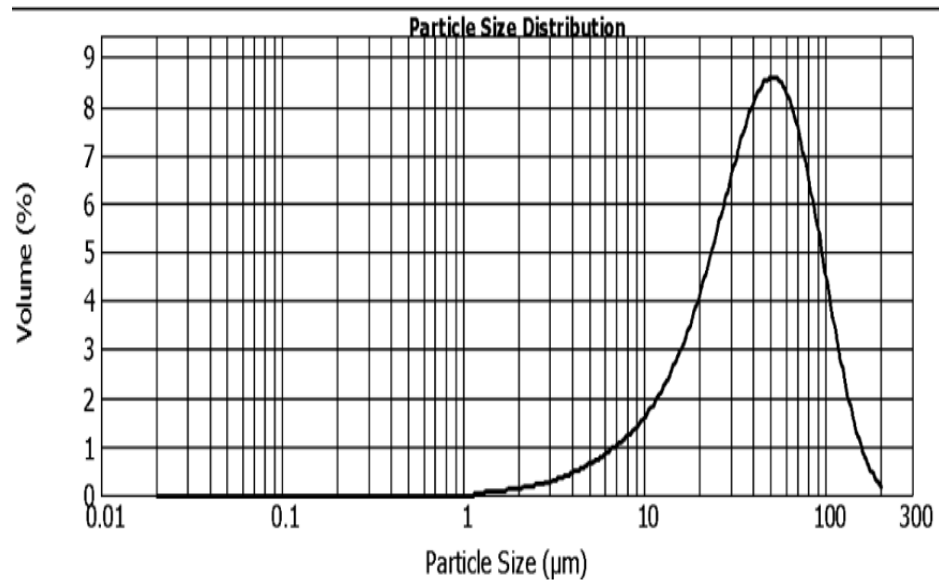


Figure 4.4 Particle Size Distribution of Graphene

4.3. Fabrication of AZ91 and 5% Graphene

Figure 4.5 show the green body of AZ91 reinforced graphene after compaction and sintering. The powder will be compacted and sinter until getting solid state. The first objective of this study were achieved which is to fabricate AZ91 reinforced 5% Graphene



Figure 4.5 An AZ91 sample reinforced graphene after compaction



Figure 4.6 An AZ91 sample reinforced graphene after sintering (a) close up view
(b) general view

4.4. XRD

The structural characteristics of the AZ91 magnesium alloy reinforced with graphene were investigated using X-ray diffraction (XRD). Figure 4.6 depicts the XRD diffraction pattern for AZ91 reinforced graphene. The obtained diffraction pattern revealed distinct peaks corresponding to different elements present in the composite material. Notably, the peaks associated with magnesium (Mg), zinc (Zn), aluminium (Al), and graphite were clearly visible.

Among these elements, magnesium had the highest peak intensity, indicating a relatively higher Mg concentration in the sample. This finding is consistent with the theoretical composition of the AZ91 alloy, in which magnesium is a major constituent. The presence of a prominent Mg peak in the XRD pattern indicates that graphene reinforcement was successfully incorporated into the AZ91 alloy matrix.

The presence of graphene reinforcement in the composite material can have a number of positive effects, including improved mechanical properties and performance. The XRD results support the idea that the graphene reinforcement has effectively integrated with the AZ91 matrix, influencing the alloy's structural properties.

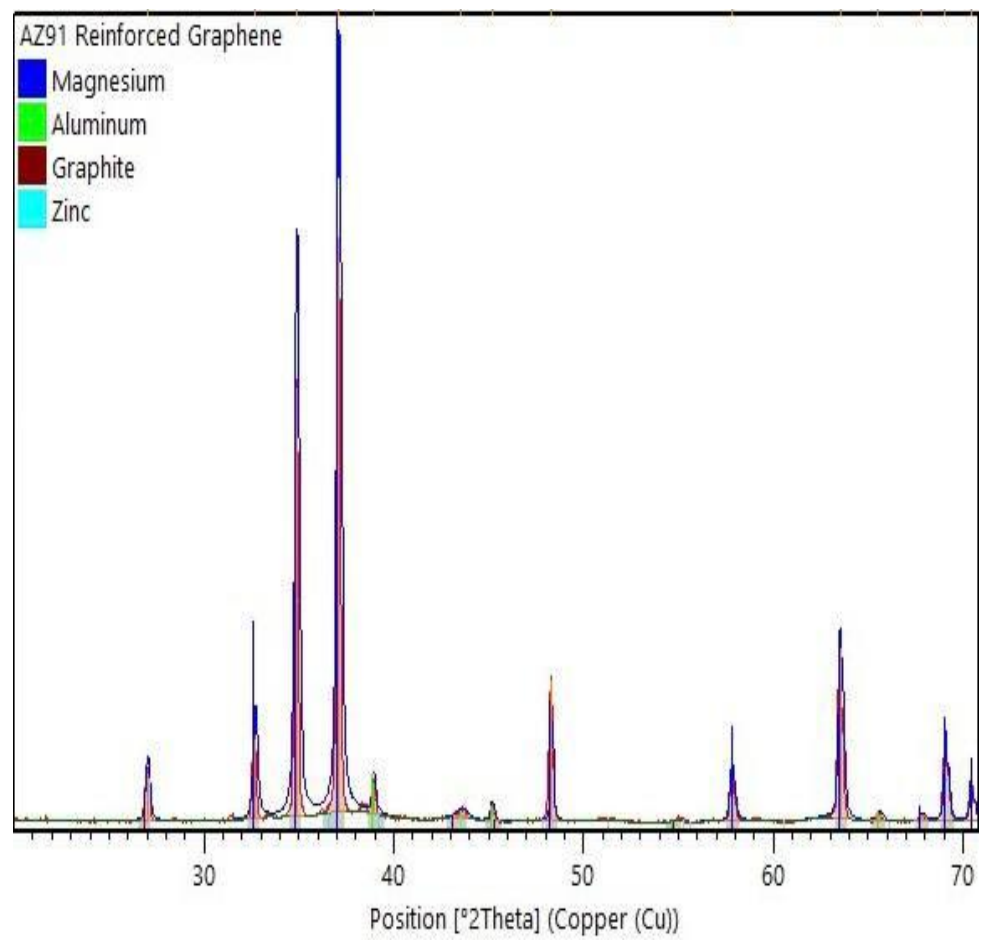


Figure 4.7 XRD diffraction pattern for Mg, Zn, Al and Graphite

4.5. Metallographic analysis

In this section, optical microscopy analysis was conducted on the cross-sectional area of an AZ91 magnesium alloy reinforced with 5% graphene at three different temperatures: 200°C (low), 300°C (average), and 400°C (high). The focus of this section was to examine the influence of temperature on the porosity of the composite material.

Furthermore, figure 4.8 until 4.10 show that the size of the pores changes with temperature. The pore size appears slightly larger at 200°C than at 300°C. However, at 400°C, the pore size is at its smallest. According to this observation, the optimum sintering temperature for achieving minimal porosity in the AZ91 reinforced with 5% graphene is 400°C.

The achievement of the lowest porosity at 400°C corresponds to the requirements of the intended application, which seeks to use a material with the fewest pores. It indicates that the material achieves a higher level of densification at this temperature, resulting in improved mechanical properties and overall performance.

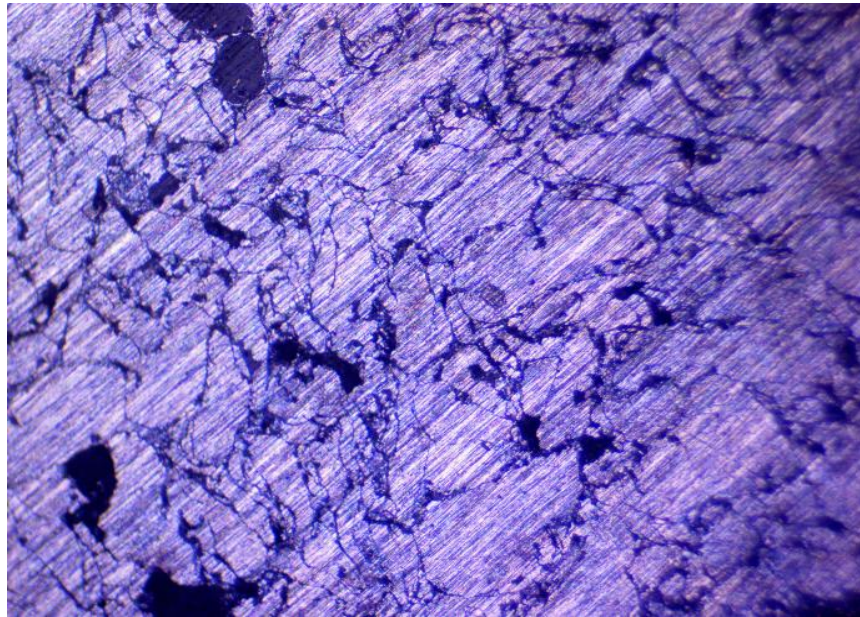


Figure 4.8 Microstructure of Magnesium composite at 200°C(10X)

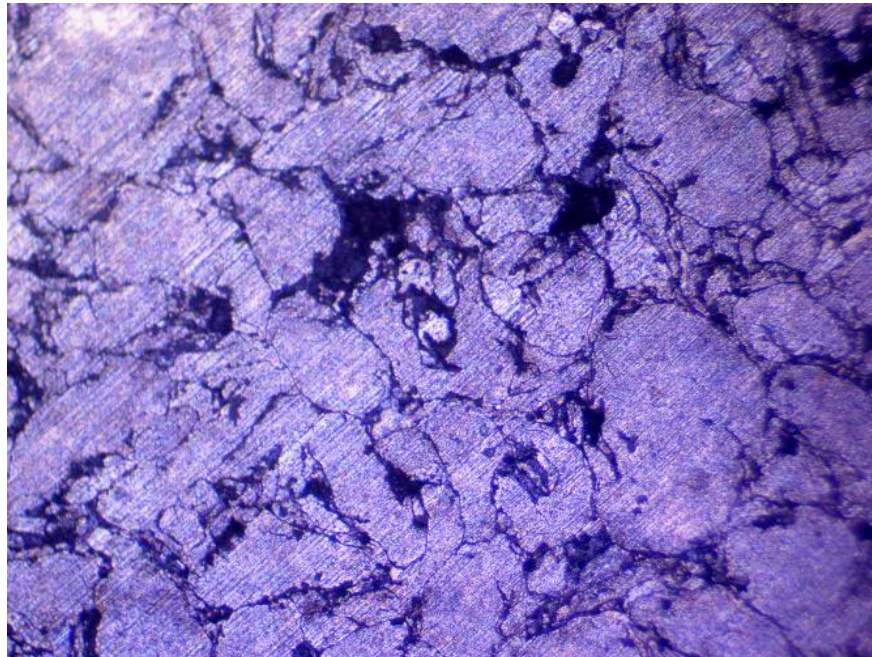


Figure 4.9 Microstructure of Magnesium composite at 300°C(10X)

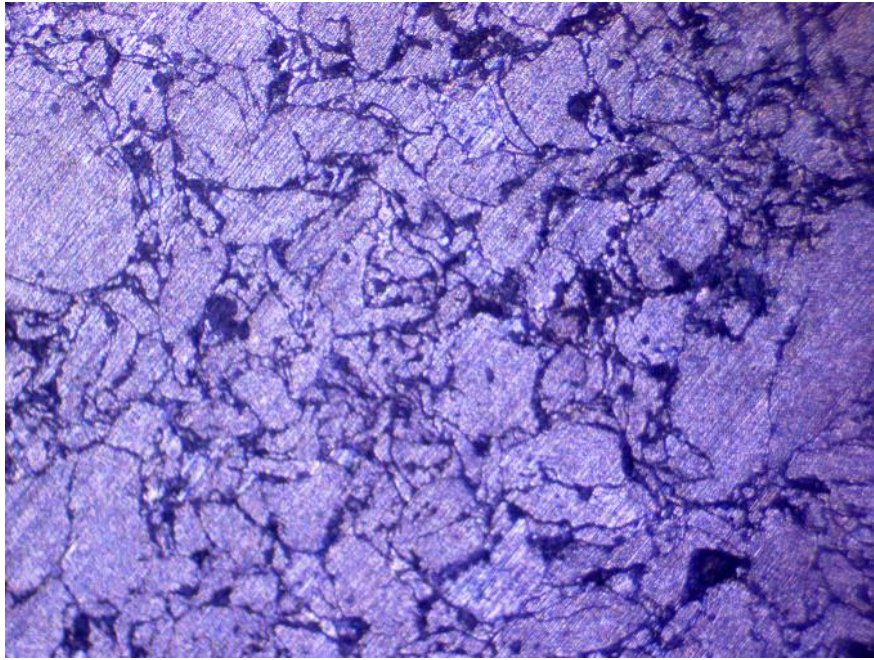


Figure 4.10 Microstructure of Magnesium composite at 400°C(10X)

4.6. Mechanical Properties of Magnesium Composite

4.6.1. Rockwell Hardness Test

The hardness values of AZ91 magnesium alloy reinforced with 5 wt% graphene are shown in Table 4.1. The addition of graphene to the AZ91 matrix has clearly resulted in an increase in hardness. This enhancement is due to the grain refinement effect caused by graphene nanoplatelets (GNPs) acting as a reinforcing phase during the matrix's heterogeneous nucleation. According to the Hall-Petch relationship, which describes the strengthening

effect of grain refinement, the effective addition of GNPs plays an important role in enhancing the material's strength (Zhao et al., 2022).

Furthermore, the results in the table show that the hardness increases as the temperature rises. Notably, the sample sintered at 400°C has the highest hardness value (97.2 HrF). This finding implies that 400°C is the optimum sintering temperature for achieving the desired hardness properties in AZ91 reinforced with graphene.

Table 4.1 Hardness of AZ91+5 wt% Graphene

TEMPERATURE (°C)	1 ST INDENT (HrF)	2 ND INDENT (HrF)	3 RD INDENT (HrF)	AVERAGE (HrF)
200°C	88.6	89.9	87.1	88.5
250°C	88.3	89.2	93.9	90.5
300°C	90.4	100.1	96.5	95.7
350°C	91.7	99.4	97.2	96.1
400°C	100.4	94.9	96.2	97.2

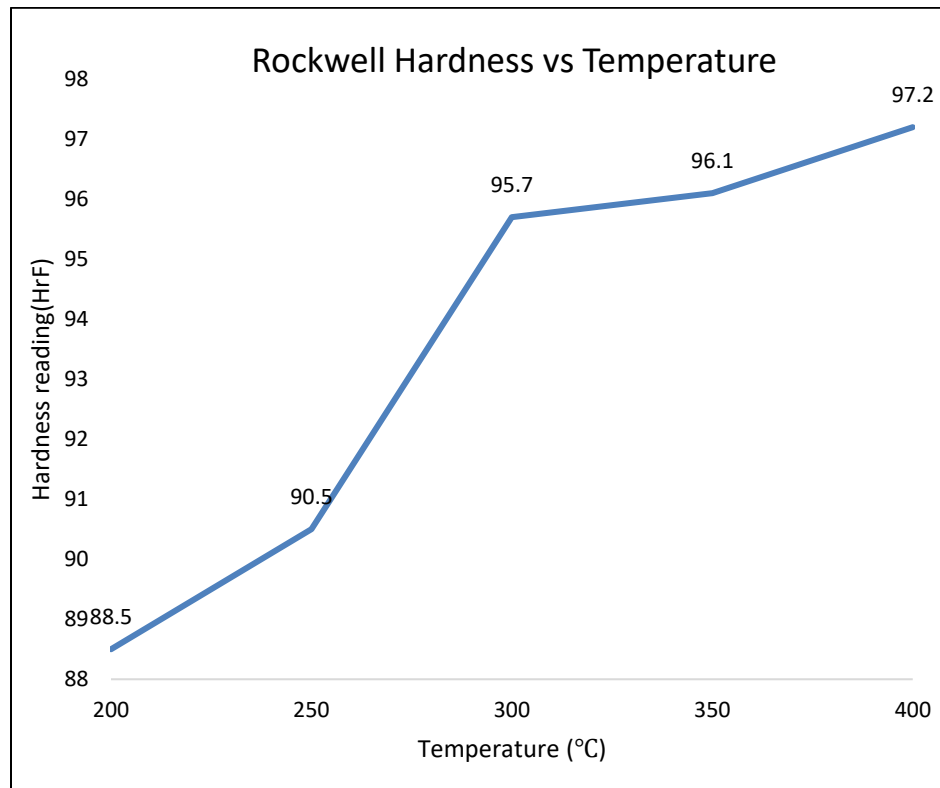


Figure 4.11 Rockwell Hardness of AZ91 reinforced 5 wt% Graphene

Figure 4.11 shows the hardness graph of AZ91 reinforced with graphene, demonstrating the variation in hardness at different temperatures. The graph depicts an increasing trend, beginning with a relatively low hardness value at 200°C and steadily increasing to its maximum at 400°C. This observation supports the notion that the optimal sintering temperature for achieving the desired hardness is indeed 400°C.

4.6.2. Density and Porosity of AZ91+5 wt% Graphene

The bulk density of a material is the mass per unit volume of the material. It is a measure of how tightly the material's particles are packed, whereas the relative density of a material is the ratio of its density to the density of water. It is a measure of how dense the material is in comparison to water. The Bulk density of material can be calculated using equation 3.2, and the relative density can be calculated using equation 3.3. In the experimental results, it is observed that the calculated bulk density for all the produced samples is slightly lower than the theoretical density.

Upon analyzing Table 4.2, it becomes clear that as the sintering temperature was raised, the bulk density of the samples also experienced an increase. This outcome can be attributed to the reinforcing impact of graphene particles, which acted as connectors between the AZ91 particles, resulting in a denser and more compact material structure. The heightened bulk density was a direct outcome of the graphene's presence, effectively occupying the gaps and spaces within the AZ91 matrix. At a sintering temperature of 400°C, the sample shows the highest bulk density with a measurement of 1.626 g/cm³. Additionally, the substance has a high density compared to water with a relative density of 81.6% at this temperature. These findings show the effectiveness of the sintering process and the addition of graphene in producing a very dense material.

The material is densely packed, as indicated by the high bulk density and relative density values at 400 °C, which may have a significant impact on its mechanical properties and potential applications. Strength, stiffness, and overall performance are frequently enhanced in a dense material.

Table 4.2 Bulk Density and Relative Density of AZ91+5 wt% Graphene

Temperature(°C)	Dry Weight(g)	Wet Weight (g)	Soak weight(g)	Bulk Density (g/cm ³)	Relative Density(%)
200°C	1.98	0.83	2.09	1.571	78.8
250°C	1.99	0.86	2.12	1.579	79.2
300°C	2.01	0.88	2.14	1.595	80.0
350°C	1.97	0.86	2.09	1.601	80.3
400°C	2.00	0.87	2.10	1.626	81.6

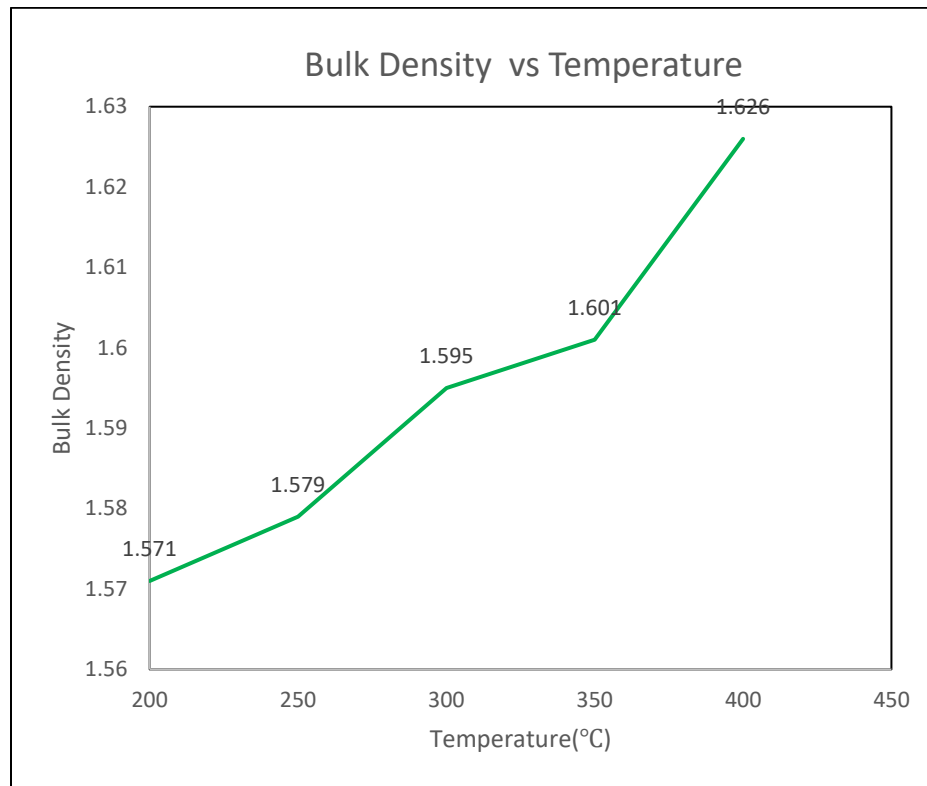


Figure 4.12 Bulk Density of AZ91 reinforced 5 wt% Graphene

The graph of the bulk density of AZ91 magnesium alloy reinforced with graphene at various sintering temperatures is shown in Figure 4.12. The graph gives a visual representation of how bulk density changes as temperature changes. The bulk density initially starts at a relatively low value at 200 °C, as can be seen by analysing the graph. The bulk density gradually rises to 250 °C, 300 °C, and 350 °C as the sintering temperature rises. However, it peaks on the graph at 400 °C, where it reaches its maximum value. This pattern suggests that the sintering temperature has a significant impact on the composite material's bulk density.

The peak at 400 °C indicates that this temperature is ideal for achieving the highest bulk density in the graphene-reinforced AZ91. This result is consistent with earlier discussions emphasizing the effect of graphene reinforcement and sintering temperature on the material's density.

The graph shows the relationship between sintering temperature and bulk density, making it possible to determine the ideal temperature for maximising the density of the composite material. The graph's findings can be helpful in optimising the manufacturing process and modifying the material's properties for particular applications where density is crucial.

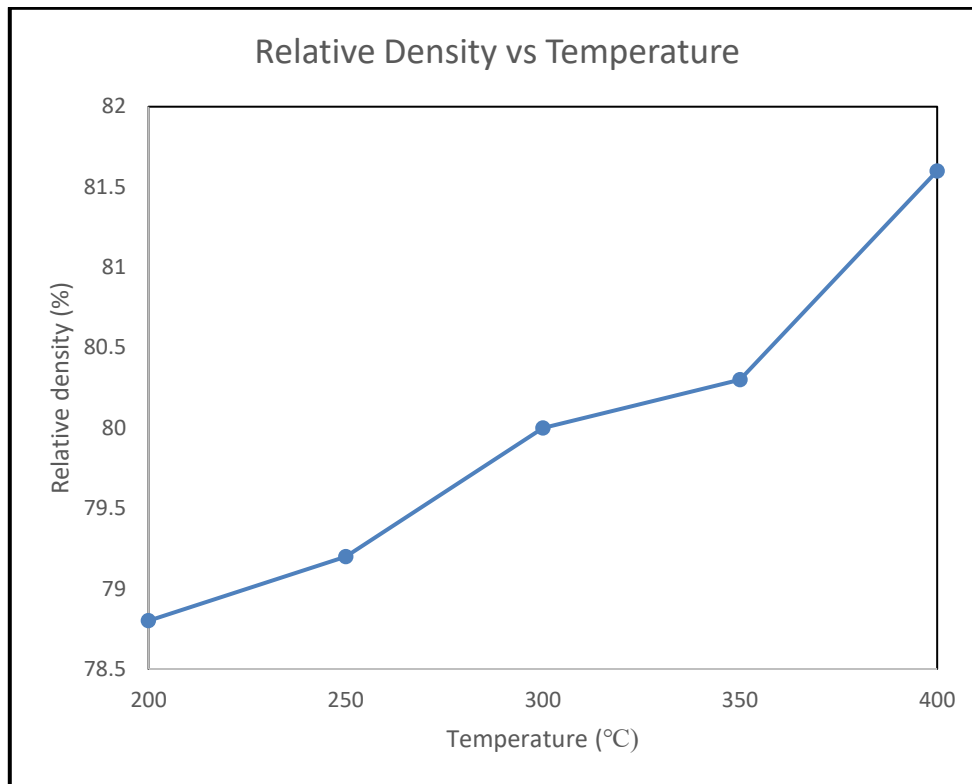


Figure 4.13 Relative Density of AZ91 reinforced 5 wt% Graphene

Figure 4.13 presents the graph of the relative density of AZ91 magnesium alloy reinforced with graphene at different sintering temperatures.. The graph gives a visual representation of how relative density changes as temperature changes.

Analyzing the graph, it is observed that the relative density increases across the different sintering temperatures. The relative density values begin at a lower value of 78.8% and gradually increase as the temperature rises. The trend continues until it reaches a higher relative density value of 81.6% at 400°C.

The graph's increasing pattern indicates that the sintering temperature has a significant influence on the relative density of the composite material. The material undergoes densification as the temperature rises, resulting in a higher relative density value. Particle rearrangement, compaction, and reduction in voids and porosity within the material are all factors that contribute to densification.

The observed increase in relative density is a positive result, indicating a higher density than water. A higher relative density indicates a more compact and densely packed material, which can be beneficial in applications where density is critical.

The graph depicts the relationship between sintering temperature and relative density in a clear visual manner, allowing for the identification of temperature ranges that yield higher relative density values. This data can be used to improve the manufacturing process and tailor the material's properties to specific application requirements.

The porosity results for the entire AZ91 magnesium alloy reinforced with 5 wt% graphene sample at various sintering temperatures are summarised in Table 4.3. Porosity is the presence of voids or empty spaces within a material, and lower porosity indicates a higher density and fewer voids.

The table clearly shows that as the sintering temperature increases, the porosity of the graphene-reinforced AZ91 decreases. Higher temperatures appear to promote better particle packing and bonding, resulting in a denser material with fewer voids. The sample had the lowest porosity of 18.4% when sintering at 400°C. Low porosity values are significant because they indicate that the material has fewer voids, which improves structural integrity and mechanical properties. Low porosity is especially important when producing bone implants with a low risk of infection. With fewer voids available, bacteria have fewer places to colonise, lowering the likelihood of infection.

The relationship between high density, low porosity, and the application goal of producing bone implants with low infection risk emphasises the significance of selecting an optimal sintering temperature. The optimum sintering temperature is 400°C due to its ability to yield a material with high density and low porosity, which aligns with the desired properties for bone implant applications.

Table 4.3 Total porosity of AZ91+5 wt% Graphene

Temperature	Relative Density(%)	Total Porosity(%)
200	78.8	21.2
250	79.2	20.8
300	80.0	20
350	80.3	19.7
400	81.6	18.4

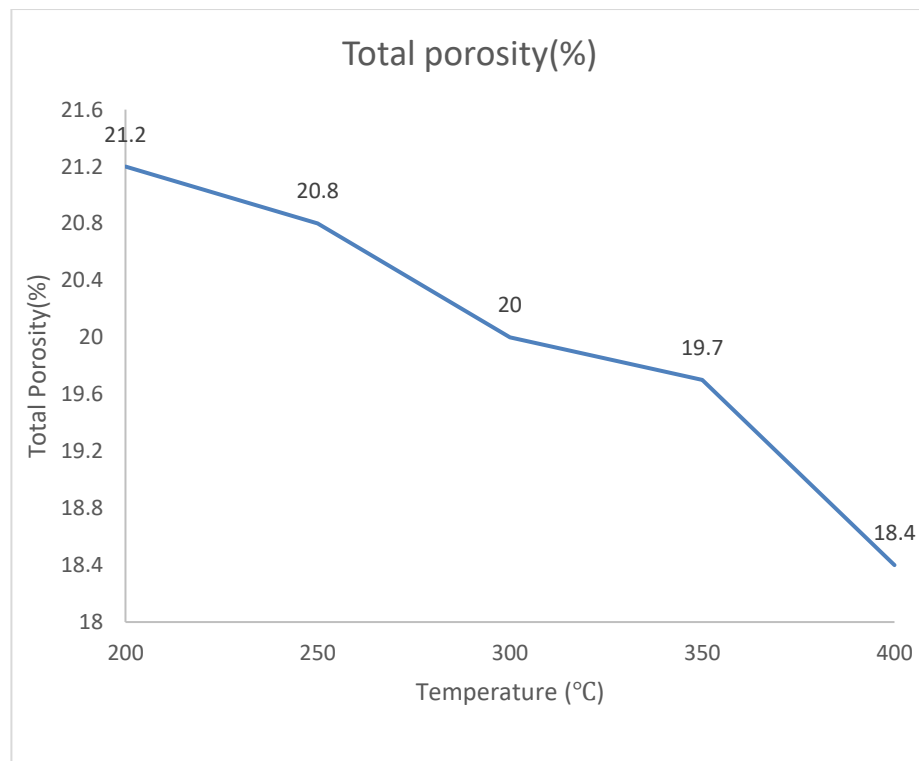


Figure 4.14 Total Porosity of AZ91 reinforced 5 wt% Graphene

Figure 4.14 shows the porosity of an AZ91 magnesium alloy reinforced with graphene at various sintering temperatures. The graph depicts how the porosity changes as the temperature rises and falls.

Examining the graph, it is clear that the porosity decreases as the sintering temperature rises. The initial porosity value is relatively higher at the lowest temperature studied, and the porosity gradually decreases as the temperature rises. The trend continues until the lowest porosity value of 18.4% is reached at 400°C. This pattern indicates that the sintering temperature is important in reducing porosity within the material. Higher temperatures improve particle packing, bonding, and void formation, resulting in a denser and more compact material with lower porosity.

The graph shows decreasing porosity, which indicates a material with fewer voids and improved structural integrity. Lower porosity is frequently associated with improved mechanical properties such as strength and toughness. The graph also depicts the relationship between sintering temperature and porosity, allowing the identification of temperature ranges that result in lower porosity values.

An inverse relationship between the two properties can be seen in the results for the density and porosity of the AZ91 magnesium alloy reinforced with graphene. A more densely packed material was indicated by higher density values, which were a result of raising the sintering temperature. As the sintering temperature increased, the porosity also decreased a sign that the amount of voids and empty spaces in the material had decreased. These results highlight the critical role that temperature control plays in obtaining materials with higher density and lower porosity, which are desirable for improved mechanical properties and applications that call for a more compact and solid structure.

CHAPTER 5

CONCLUSION AND RECOMMENDATION

5.1. Conclusion

Based on the results and data analysis of this research, two key conclusions can be drawn. Firstly, powder metallurgy is a suitable method for fabricating AZ91 reinforced with graphene, enabling control over the pore structure and providing enhanced reinforcement for natural bone. Secondly, the investigation of different sintering temperatures revealed that the optimum sintering temperature for fabricating AZ91 reinforced with graphene is 400°C.

At this optimal sintering temperature, significant improvements were observed in the material properties. The composite exhibited the highest hardness value of 97.2 HrF, indicating enhanced mechanical strength. Additionally, the bulk density reached its peak value of 1.626 g/cm³, indicating a denser material with improved structural integrity. Furthermore, the lowest porosity value of 18.4% was achieved, indicating a reduced presence of voids and improved material quality.

These findings highlight the critical role of the sintering temperature in determining the final properties of the composite material. By optimizing the sintering temperature, the grain refinement, particle distribution, and interfacial bonding between the graphene reinforcement and the AZ91 matrix were optimized, resulting in improved mechanical strength, density, and porosity. These insights contribute to the understanding of the fabrication process and provide practical guidance for achieving desirable material properties in AZ91 reinforced with graphene.

Overall, the determination of the optimum sintering temperature is a significant finding of this research, as it informs the fabrication process and helps in the development of high-performance materials for various applications.

5.2. Recommendation

This study's advancement was critical to the emergence of magnesium composite in biomedical applications and a better understanding for future studies. Powder metallurgy procedures must be followed carefully because they affect the density and porosity of the sample.

Proper compaction of the AZ91/graphene powder is essential for achieving high-density samples with minimal defects. Optimize the compaction parameters, including pressure, dwell time, and temperature (if applicable), to ensure uniform compaction and prevent any porosity or voids in the composite. Implement suitable die design and consider the use of lubricants, if required, to prevent sticking and facilitate smooth compaction. Carefully select the sintering temperature, taking into account the melting point of the components, particularly the AZ91 matrix and graphene reinforcement. Consider the temperature range that ensures good sintering and densification without causing any adverse reactions or degradation of the composite.

Some recommendation for future works would be as follows:

1. Studies varying composition of Magnesium with weight percent of Graphene.
2. Varying processing parameter such as milling time.
3. Investigate surface modification technique such as plasma treatment to enhance the bonding between graphene and the AZ91 matrix.

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APPENDIX

Calculation :

Temperature(°C)	Dry Weight(g)	Wet Weight (g)	Soak weight(g)
200°C	1.98	0.85	2.09
250°C	1.99	0.86	2.12
300°C	2.01	0.88	2.14
350°C	1.97	0.86	2.09
400°C	2.00	0.87	2.10

$$\text{Bulk Density} = \frac{\text{Weight of dry sample}}{W_c - W_b} \times \rho_{\text{water}}$$

Sample 1 : 200 °C

$$\begin{aligned}\text{Bulk Density} &= \frac{1.98}{2.09 - 0.85} \times 1.0 \\ &= 1.571\end{aligned}$$

Sample 2 : 250 °C

$$\begin{aligned}\text{Bulk Density} &= \frac{1.99}{2.12 - 0.86} \times 1.0 \\ &= 1.579\end{aligned}$$

Sample 3 : 300 °C

$$\begin{aligned}\text{Bulk Density} &= \frac{2.01}{2,12-0.86} \times 1.0 \\ &= 1.595\end{aligned}$$

Sample 4 : 350 °C

$$\begin{aligned}\text{Bulk Density} &= \frac{1.97}{2,09-0.86} \times 1.0 \\ &= 1.601\end{aligned}$$

Sample 5 : 400 °C

$$\begin{aligned}\text{Bulk Density} &= \frac{2.00}{2,10-0.87} \times 1.0 \\ &= 1.626\end{aligned}$$

Temperature	Bulk Density (g/cm ³)
200	1.571
250	1.579
300	1.595
350	1.601
400	1.626

$$\text{Relative Density} = \frac{\text{Bulk Density}}{\text{Theoretical Density}} \times 100\%$$

Density Mg: 1.738 g cm⁻³

Density Al :2.7 g cm⁻³

Density Zn: 7.13 g cm⁻³

Density Graphene : 2.3 g cm⁻³

Theoretical Density of AZ91+5 wt% Graphene

$$= (0.9)(1.738) + (0.01)(2.7) + (0.05)(2.3)$$

$$= 1.9935$$

Sample 1 : 200 °C

$$\text{Relative Density} = \frac{1.571}{1.9935} \times 100 \%$$

$$= 78.8\%$$

Sample 2: 250 °C

$$\text{Relative Density} = \frac{1.579}{1.9935} \times 100 \%$$

$$= 79.2\%$$

Sample 3 : 300 °C

$$\begin{aligned}\text{Relative Density} &= \frac{1.595}{1.9935} \times 100 \% \\ &= 80.0\%\end{aligned}$$

Sample 4 : 350 °C

$$\begin{aligned}\text{Relative Density} &= \frac{1.601}{1.995} \times 100 \% \\ &= 80.3\%\end{aligned}$$

Sample 5 : 400 °C

$$\begin{aligned}\text{Relative Density} &= \frac{1.626}{1.9935} \times 100 \% \\ &= 81.6\%\end{aligned}$$

Temperature	Relative Density(%)
200	78.8
250	79.2
300	80.0
350	80.3
400	81.6

Total Porosity= 100 % - Relative Density

Sample 1: 200 °C

Total porosity = 100%- 78.8%
= 21.2%

Sample 2 : 250 °C

Total porosity = 100%- 79.2%
= 20.8%

Sample 3 : 300 °C

Total porosity = 100%- 80.0%
= 20 %

Sample 4 : 350 °C

Total porosity = 100%- 80.3%
= 19.7%

Sample 5 : 400 °C

Total porosity = 100%- 81.6%

= 18.4%

Temperature	Relative Density(%)	Total Porosity(%)
200	78.8	21.2
250	79.2	20.8
300	80.0	20
350	80.3	19.7
400	81.6	18.4

CURRICULUM VITAE

A. Personal profile

Full name Nur Hazimah Binti Sadon
National IC no 010402 09 0088
Birth date 2nd April 2001
Citizenship MALAYSIA
Place of birth Perlis, Malaysia
Gender Female
Correspondence address Kampung Batas Lintang
02700 Simpang Empat, Perlis

Telephone no. (H) -
Telephone no. (HP) 6017 4181458
Email address nurhazimah731@gmail.com



B. Hobbies and interests

I enjoy traveling to new and exciting destinations. I love engaging in extreme and adventure activities, like bungee jumping and flying fox, which give me an adrenaline rush. I also have a passion for tutoring and teaching, as I find it fulfilling to share knowledge and inspire learning. I like playing sports like futsal, netball, and handball to stay active. I enjoy meeting new people and volunteering with local organizations. I love learning new things, whether it's general knowledge or challenging IQ games.

I am very fluent in written and spoken Malay and English and have basic knowledge of Arabic.

C. Academic qualifications

Degree	Area	Institution	Year awarded
B.Sc. (Hons.)	Science Physics	Universiti Teknologi MARA, Malaysia	2024
Matriculation	Science Stream	Kolej Matrikulasi Perlis, Arau	2020
S.P.M	Science Stream	SMKA(P) Alawiyah Kangar, Perlis	2018

D. Work experience

Post	Place	Year
Addmath Online Tutor	Tutorkami.com	2022

E. Related experience

Post	Place	Year
Vice Secretary	Physics Society (PhySo), Universiti Teknologi MARA, Arau	2021-2022
Facilitator	Motivational Programme, Cocurriculum, Universiti Teknologi MARA, Arau	2021-2022